

Darpen Bhandari

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Portfolio: darpenbhandari.com | Work Authorization: Valid German residence permit | Available Immediately

Summary

A Data Engineer with 5 years building Azure native data platforms across finance and healthcare, including team lead experience managing a 6 engineer squad. Designed medallion lakehouses (ADF, Databricks), automated 10+ hours of weekly reconciliation and maintained 100% SLA compliance across 22 portfolio applications at Hexaware.

Professional Experience

Data and System Analytics Associate (Part-Time)

12/2025 – 05/2026

Universitätsklinikum Heidelberg, Heidelberg, Germany

- Reduced 20% of manual reporting effort by designing Python based REST API ingestion pipelines between qTest and Matrix42, standardising incident, service request and test execution of data into a single weekly reporting layer that delivered in close collaboration with IT service management teams who defined the reporting requirements.
- Developed end-to-end ADF pipeline design, parameterised triggers, incremental load patterns and ADLS integration enabling scheduled SLA breach monitoring across IT service workflows.
- Improved Power BI data reliability by implementing dbt staging, intermediate and mart models with automated schema, relationship and null value tests, cutting ad-hoc data quality issues in operational reports used by IT operations and service delivery teams.
- Prevented data regressions from reaching production by introducing CI/CD gates via GitHub Action running dbt tests, SQL checks and pipeline config validation on every PR before deployment.

Senior Software Engineer

03/2019 – 10/2023

Hexaware Technologies Pvt. Ltd., Navi Mumbai, India

- Led a team of 6 data engineers delivering pipeline and reporting solutions across 22 US financial portfolio applications and co-ordinating sprint planning, code reviews and production releases in a weekly Agile delivery cycle aligned with month end reporting deadlines.
- Maintained 100% SLA compliance across all portfolio applications by architecting ADF pipelines with incremental load logic, failure alerting and automated retries, working alongside the DevOps team on deployment and monitoring.
- Saved almost 10 hours of weekly reconciliation effort by replacing manual Excel based AUM and performance reporting with fully automated Python and SQL pipelines on Azure freeing the finance analyst team to focus on analysis over data preparation.
- Designed a medallion architecture (Bronze,Silver,Gold) on Parquet based ADLS and modelled outputs as star schema facts and dimensions, embedding automated data quality checks, null detection, duplicate flagging and reconciliation gaps that alerted directly into portfolio pipelines to reduce month end investigation effort.

Key Projects

Smart Energy Forecasting Pipeline- Master's Thesis

04/2025 – 09/2025

SRH Hochschule Heidelberg, Heidelberg, Germany

- Identified 12% potential energy savings across building profiles by designing an end-to-end pipeline ingesting electricity, solar, weather and cost datasets into a Bronze,Silver,Gold medallion lakehouse on ADLS.
- Built dbt models for energy consumption, renewable generation and KPI mart layers with automated tests for missing timestamps, duplicate records and invalid consumption values.
- Delivered modular transformation logic with parameterised aggregation windows, feeding downstream forecasting models and a Power BI dashboard for building level energy analysis.

Stroke Risk Analytics Dashboard

04/2025 – 06/2025

Personal Project - GitHub

- Cleaned and modelled 5,110 healthcare records using Python and SQL, structuring dbt staging and mart models for demographic, BMI, smoking and stroke risk analysis.
- Delivered an interactive Power BI dashboard with KPI cards, slicers and risk breakdowns, enabling faster exploratory analysis across patient cohorts.
- Applied validation checks for null values, duplicate patient records and category consistency, ensuring dataset integrity before any analytical output was produced.

Education

M.Sc. Applied Data Science and Analytics

10/2023 – 09/2025

SRH Hochschule Heidelberg, Heidelberg, Germany

Technical Skills

Programming & Querying: Python, SQL, PySpark (batch transformations), Pandas, NumPy, REST APIs, DAX

Cloud & Data Engineering: Azure Data Factory, Azure Data Lake Storage, Azure SQL, Azure Synapse Analytics (familiar), Azure DevOps, Databricks

ETL/ELT & Orchestration: Batch Pipelines, Incremental Loads, ADF Triggers & Parameterisation, Pipeline Monitoring, SLA Checks, Reconciliation

Lakehouse & Storage: Parquet, Delta Lake, Medallion Architecture (Bronze/Silver/Gold), Data Warehousing

Data Transformation: dbt (Staging, Intermediate & Mart Models), dbt Tests (Schema, Relationship, Duplicate), Power Query

Data Quality: Automated Validation Checks, Null Detection, Reconciliation Gap Alerts, SLA Monitoring

DevOps & CI/CD: Git, GitHub Actions, Docker, Pipeline Version Control

Delivery & Collaboration: Agile (Scrum), Sprint Planning, Code Reviews, Cross functional Stakeholder Engagement, Team Leadership (6 engineers)

Data Modelling: Star Schema, Fact & Dimension Modelling, Reporting Marts, Power BI Semantic Models

BI & Visualisation: Power BI, DAX, Tableau

Domain Expertise: Financial portfolio data (AUM, holdings, exposure, trade reconciliation, performance reporting) ; healthcare analytics (patient risk stratification, clinical KPIs) ; IT service operations (incident management, SLA compliance)

Languages

English: C1 (Professional working proficiency) | **German:** B1 (actively improving, B2 target)

Certificates

Microsoft Azure Fundamentals (AZ-900) | **Databricks** for Data Engineering | **NVIDIA** Deep Learning Fundamentals